

NeuroFC — AI Flight Controller

Autonomous Edge-AI UAV Brain | Hovernest's Core Technology

OVERVIEW

NeuroFC is Hovernest's proprietary AI-native flight control system — the intelligent brain behind every autonomous UAV mission. Designed for real-time onboard inference, adaptive decision-making, and modular integration across the entire Hovernest platform ecosystem, with zero dependency on cloud compute for mission-critical operations.

TECHNICAL SPECIFICATIONS

Compute Platform	AI Edge Processor (NVIDIA Jetson / equivalent)
AI Inference Engine	Onboard real-time neural network
IMU	9-DOF, 1000 Hz update rate
GPS Module	Multi-constellation RTK GNSS
Vision Module	Stereo depth + monocular camera
Connectivity	4G/5G LTE + RF Telemetry (915 MHz / 2.4 GHz)
Protocol	MAVLink 2.0 + custom NeuroFC API
Swarm Readiness	Multi-agent coordination (upcoming release)
Operating Temperature	-20°C to +60°C
Compliance	DGCA-Compliant Architecture

KEY FEATURES

- ▶ Onboard YOLOv8-optimised real-time object detection and classification
- ▶ Dynamic path optimisation with AI-driven obstacle avoidance
- ▶ Adaptive flight stabilisation via live multi-sensor fusion
- ▶ Predictive maintenance alerts through anomaly detection
- ▶ Open integration: PX4, ArduPilot, and proprietary stack support
- ▶ Swarm coordination framework — multi-drone formation (future release)

USE CASES

- ▶ Fully Autonomous Medical Delivery
- ▶ Smart Agriculture & Precision Spraying
- ▶ Defense, Surveillance & Reconnaissance
- ▶ Autonomous Logistics Networks
- ▶ AI Research & Development Platform
- ▶ Drone-as-a-Service (DaaS) Deployment